



- 5 An engineer is comparing the tensile strengths of steel rods made from two machines, A and B . The engineer randomly selects 8 rods from machine A and 6 rods from machine B . The tensile strengths, in appropriate units, are given in the following table.

Machine <i>A</i>	402	403	415	412	409	407	406	410
Machine <i>B</i>	401	398	395	397	410	405		

You should assume that the two distributions are normal and have the same population variance.

Use a t -test at the 5% significance level to test whether there is any difference in the mean tensile strengths of steel rods from the two machines. [9]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

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